Visual Permeability-Driven Generation of Support-Free Stochastic Porous Structures

Abstract

Porous structures are ubiquitous in nature, engineering, and landscape design, prized for their lightweight characteristics and rich geometric expressiveness. However, fabricating such organically irregular and visually permeable structures via natural weathering or subtractive manufacturing is severely limited. While additive manufacturing (AM) enables fabrication for complex geometries, manual modeling demands substantial design expertise and intensive effort, and the internal supports required during printing compromise morphology and are difficult to remove. Furthermore, quantitative metrics to evaluate visual permeability (VP) are critically lacking. To generate stochastic porous structures that are highly visually permeable, explicitly controllable, and support-free, we propose a novel generation and optimization framework tailored for AM. First, a quantitative VP metric drives a graph-based connectivity optimization to maximize aesthetic transparency and structural sparsity. Next, orthotropic Gaussian-kernel implicit fields model the continuous stochastic porous geometry, intrinsically reducing initial overhangs. Finally, a density-field optimization constrained by a differentiable layer-wise AM filter strictly enforces support-free manufacturability with minimal geometric distortion. Experimental results demonstrate that our framework automatically produces geometrically diverse, visually permeable, and support-free porous structures, bridging the gap between aesthetic procedural design and digital fabrication.

Keywords: Stochastic Porous Structures, Visual Permeability Optimization, Support-Free Additive Manufacturing.

1. Introduction

Porous structures represent an efficient form of material organization, widely observed in nature and increasingly exploited in engineering design, as shown in Figure 1(a)(b). Owing to their high specific strength, lightweight nature, and large surface-to-volume ratio, they have been extensively applied in functional domains such as tissue engineering [1, 2], catalyst supports [3], and heat exchanger design [4]. Beyond functional utility, the stochastically distributed pores and spatially connected tunnels endow these structures with rich spatial layering and unique sight-line penetration. Consequently, they are prized elements in architectural, landscape, and artistic design [5, 6, 7], exemplified by the revered Taihu stones in Eastern aesthetics (Figure 1(d)).

However, acquiring such aesthetically pleasing porous structures in the physical world is challenging. Natural masterpieces require centuries of weathering and erosion, making them exceedingly rare and non-renewable. Artificial alternatives typically rely on subtractive manufacturing, involving mechanical carving, manual drilling, or chemical surface treatments to mimic natural textures. These traditional methods are not only labor-intensive and environmentally unfriendly but also geometrically limited. Furthermore, they often produce superficial porosity with poor spatial connectivity.

Digital modeling combined with additive manufacturing (AM) offers a promising pathway to overcome these physical constraints and fabricate such intricate aesthetics (Figure 1(e)). However, manual modeling demands substantial design expertise and aesthetic sensibility, consuming significant effort. However, automating the design of such complex geometries presents a fundamental trilemma: balancing aesthetic stochasticity, explicit controllability, and support-free manufacturability. Existing solutions inevitably fail to satisfy all three. Specific design methods for Taihu stones [6, 7] often involve complex, time-consuming pipelines that neglect fabrication constraints. Performance-driven methods, such as topology optimization (TO) [8], can effectively enforce support-free constraints [9, 10] but typically yield regular, truss-like geometries that lack natural randomness. Similarly, implicit methods based on triply periodic minimal surfaces (TPMS) [11] are inherently support-free but suffer from rigid periodicity, as illustrated in Figure 1(c). While Gaussian random fields (GRF) [12, 13] excel at simulating smooth, bio-inspired stochastic morphologies, they lack quantitative aesthetic metrics and are notoriously difficult to print. Existing solutions often resort to heuristic post-processing to fix overhangs [14], which can severely distort the original morphology and compromise the intended aesthetic fidelity.

To bridge the gap between aesthetic design and digital fabrication, we require a framework that is bio-inspired, explicitly controllable, aesthetically quantifiable, and strictly fabricable. We begin by addressing the challenge of aesthetic quantification. We observe that for landscape porous structures, the key aesthetic trait is visual permeability (VP), which is not equivalent to simple statistical porosity. A structure can be highly porous yet visually occluded. Under our abstraction, VP is governed by the structural properties of pathways rather than by individual pores or tunnels. Accordingly, rather than relying on simple statistical porosity, we formulate a quantitative VP metric driven by the topological properties of internal void pathways. By conceptually evaluating the spatial continuity, straightness, length, and penetration depth of these interconnected routes, our metric provides a computable representation of aesthetic transparency and perceptual openness.

Building on this metric, we propose a fabrication-oriented framework that integrates support-free constraints into the entire generative design process. First, we employ build-direction-biased orthotropic kernels adapted from [12]. By intentionally elongating the kernels along the printing axis, this fabrication-aware parameterization intrinsically reduces overhang regions during geometric initialization without compromising organic smoothness. Subsequently, we apply a continuous density-field optimization with differentiable overhang-angle constraints, transforming the stochastic geometry into a globally support-free structure while minimizing deviation from the intended design.

[2]Image sources — (d): Photo by Gisling, Wikimedia Commons, CC BY-SA

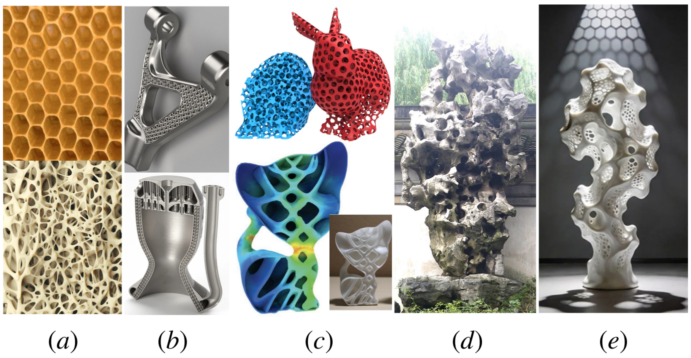


Figure 1: **Porous structures in nature, industry, and landscape. (a)** Natural examples: honeycombs and trabecular bone. **(b)** Engineering applications for lightweight efficiency. **(c)** Computational structures (GRF [12] and TPMS [11]). **(d)** Traditional Taihu stones exhibiting high visual permeability. **(e)** Additively manufactured Taihu-like structures[2].

In summary, the main contributions of this work are threefold:

- A fabrication-oriented framework for generating stochastic porous structures that are strictly support-free under a prescribed build direction.
- An integrated implicit modeling and optimization pipeline that unifies stochastic geometry generation, metric-driven connectivity optimization, and manufacturability enforcement.
- A decoupled connectivity optimization strategy driven by a quantitative visual permeability metric, enabling enhanced perceptual openness with controlled geometric complexity.

2. Related Work

In this section, we review related literature categorized by the generation logic and their handling of additive manufacturing constraints. We distinguish between optimization-based methods, periodic procedural modeling, and stochastic modeling approaches, highlighting the specific gaps regarding support-free manufacturability and aesthetic stochasticity.

2.1. Manufacturability-Constrained Topology Optimization

Topology optimization generates structures by optimizing material distribution under physical boundary conditions, naturally favoring lightweight and stiff porous patterns [8, 15].

To address AM limitations, classical density-based, level-set, and phase-field formulations have been extended to incorporate overhang constraints. A primary strategy involves modifying the projection or filtering schemes during optimization. Gaynor et al. [9] pioneered a support-free constraint integrated directly into the projection process, preventing the formation of unprintable features. Qian et al. [16] proposed a differentiable density gradient constraint utilizing Heaviside projection to simultaneously penalize undercuts and enforce minimum overhang angles. Similarly, Garaigordobil et al. [17] introduced a global constraint based on the ratio of the self-supported perimeter, offering overall manufacturability control. Another line of work employs layer-wise overhang filters that emulate the layer-by-layer deposition process. Langelaar [10] developed a nonlinear AM filter that simulates the layer-by-layer manufacturing process to suppress unsupported material. This concept was further advanced by van de Ven et al. [18, 19] using continuous front propagation methods to simulate edge progression, applicable to both continuous and discrete layer-by-layer frameworks.

While these TO-based methods provide strong theoretical guarantees for mechanical performance and printability, they inherently tend to produce regular, truss-like, or beam-based geometries. Even stochastic TO formulations [20], which incorporate loading uncertainty, remain fundamentally performance-driven, prioritizing mechanical survival probability over aesthetic morphological diversity. These outcomes, driven purely by physics, lack the morphological diversity, stochasticity, and visual permeability characteristic of biological porous media. Furthermore, the computationally intensive iterative process limits their utility for interactive exploration.

2.2. Periodic and Lattice-based Porous Structures

Procedural modeling based on periodic functions or unit cells offers a geometrically explicit alternative to TO. A significant class relies on TPMS, such as Gyroid, Schwarz-P, and Diamond surfaces. Defined by mathematical implicit functions with zero mean curvature, TPMS structures are favored for their high surface-area-to-volume ratios and smooth connectivity [4, 21]. Crucially, TPMS structures are easily adjustable for support-free due to their continuous curvature. Yan et al. [11] leveraged this by modulating the amplitude of TPMS functions to ensure support-free channels for 3D printed objects. Tian et al. [22] further explored continuous grading of TPMS parameters for functionally graded materials. Beyond implicit surfaces, periodic unit cell approaches involve tiling meticulously designed microstructures into macro-scale lattices [23, 24, 25].

Although these methods can achieve support-free fabricability through parameter tuning or unit cell design, the resulting structures are inherently periodic. This regularity contrasts sharply with the stochastic, organic aesthetics found in nature, limiting their application in scenarios requiring natural visual permeability and non-uniform spatial variation.

2.3. Stochastic Porous Modeling

To capture the irregularity of natural media, stochastic modeling techniques directly construct complex morphologies using stochastic fields or tessellations. Voronoi tessellation partitions space based on seed points, producing cellular structures mimicking natural foams [26, 27, 28]. GRFs offer another powerful avenue, capable of reproducing highly organic, interconnected morphologies with continuous solid-void transitions [29, 12]. Recent works have focused on controlling the statistical properties of these fields to tailor morphological features [30, 13]. Additionally, fractal algorithms [31, 32, 33] and data-driven generative models [34, 7, 35] have been employed to synthesize intricate, multi-scale stochastic designs.

However, ensuring the manufacturability of these stochastically generated structures remains a critical gap. Unlike TPMS, the randomness of Voronoi or GRF structures leads to unpredictable local geometries that frequently violate overhang constraints. While Martinez et al. [36] proposed a specialized polyhedral distance function to guarantee support-free Voronoi cells, this solution is specific to cellular structures and restricts the geometric freedom of continuous fields. For general stochastic fields, rigid generation processes hinder the direct integration of strict manufacturing constraints, often necessitating invasive post-processing.

3.0; (e): Artwork "Silent Shadow Solitary stone" courtesy of QiThink art.

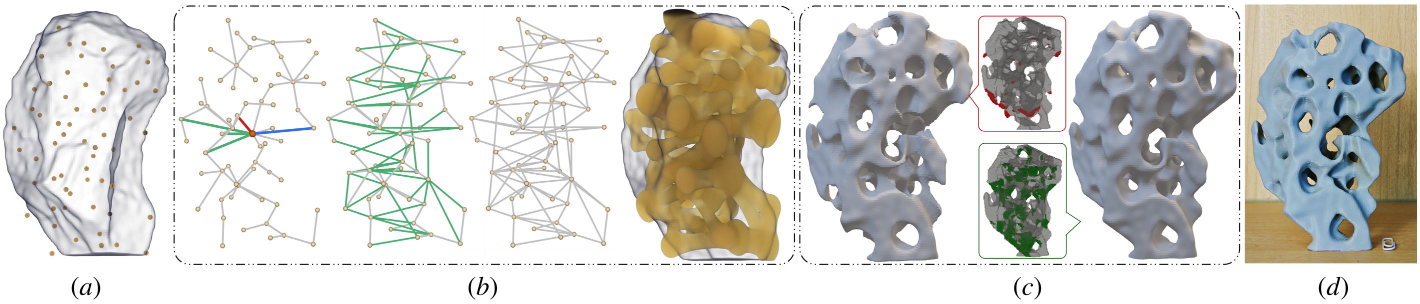


Figure 2: Pipeline for support-free porous structure generation. (a) **Initialization:** The input solid is populated with randomly sampled Gaussian kernels. (b) **Structure Generation:** Connectivity is optimized for visual permeability to generate the implicit porous geometry. Left to right: local edge operations (red replaced by blue; green directly added), accumulated edges after one iteration, the final optimized topological skeleton, and the extracted continuous isosurface. (c) **Support-Free Refinement:** Unsupported voxels (red) are pruned, while supporting voxels (green) are added to ensure manufacturability. (d) **Fabrication:** The final result, manufactured without external supports.

2.4. General Geometric Processing for Support-Free AM

Apart from structural generation, general geometric processing strategies have been developed to aid the fabrication of arbitrary 3D models. Early solutions focused on optimizing print orientation to minimize support volume [37, 38, 39]. For hollow objects, methods such as generating support-free internal contours via layer-wise slicing analysis have been proposed [14]. CurviSlicer [40] achieves this on standard 3-axis printers by computing a global deformation that allows for slightly curved slicing, effectively transforming overhangs into support-free features. For complex topologies, they proposed a volume-to-surface decomposition method [41], enabling completely support-free fabrication along curved toolpaths.

However, for highly intricate stochastic porous structures, global orientation optimization fails to eliminate internal supports completely, and multi-axis hardware lacks the universality of standard 3-axis platforms. This underscores the necessity of embedding manufacturing constraints directly into the structural generation process.

2.5. Aesthetic Quantification and Connectivity Optimization

Preserving the artistic intent of porous structures demands explicit aesthetic metrics. However, recent generative approaches for Taihu-like structures fall short in this regard. Feng et al. [6] noted that visual permeability could not be quantified within their physics-driven framework, while Deng et al. [7] merely conceptualized it to categorically annotate 2D sectional data for machine learning. Neither scalar volume ratios nor discrete 2D annotations provide an active 3D metric capable of capturing the directional, depth-aware nature of visual permeability.

Similarly, existing graph-based methods for constructing interconnected networks extract skeletons primarily for basic connectivity or fluid flow, neglecting aesthetic objectives. Minimum spanning trees (MST) [42] yield excessively sparse graphs that restrict visual pathways, whereas Morse-Smale complexes (MSC) [43, 12] generate dense, redundant connections without prioritizing sightlines. While minimum bounded degree spanning trees (MBDST) [44, 13] offer a structural compromise by limiting local connections, they remain purely topological and blind to spatial visibility. Our work bridges this gap by introducing a quantitative visual permeability metric to drive a customized bounded-degree graph optimization.

3. Methodology

This section details the computational workflow of our fabrication-oriented generative framework. The pipeline takes an input geometric domain represented by a closed triangle mesh or its signed distance field (SDF), and a prescribed build direction (the z-axis) as input, and outputs a support-free, visually permeable stochastic porous structure. We first provide an overview of the three-stage pipeline (Section 3.1). Subsequently, we introduce the quantitative VP metric (Section 3.2) and the metric-driven connectivity optimization (Section 3.3). Finally, we detail the continuous implicit modeling using orthotropic Gaussian kernels (Section 3.4) and the density-field optimization based on a layer-wise AM filter (Section 3.5).

3.1. Overview

As illustrated in Figure 2, our proposed framework consists of three interconnected stages: (1) visual permeability-driven connectivity optimization, (2) stochastic implicit pore modeling, and (3) residual overhang optimization. Unlike traditional topology optimization methods that prioritize mechanical stiffness, our approach is fundamentally driven by VP, while satisfying strict additive manufacturing constraints.

In the first stage, the design domain is initialized by randomly sampling seed points. By abstracting the porous network as a connectivity graph, where pores act as nodes and tunnels as edges, we define a novel metric to quantitatively evaluate global visual permeability. This metric guides a customized bounded-degree graph optimization to maximize visual penetrability while maintaining the necessary structural sparsity.

In the second stage, the discrete optimal graph is realized as a continuous geometry using implicit scalar fields. To inherently minimize initial printing overhangs without sacrificing organic smoothness, we model the pores using orthotropic Gaussian kernels intentionally elongated along the build direction. Continuous tunnels with elliptical cross-sections are then constructed along the optimized graph edges, seamlessly interpolating between the endpoint kernels.

In the final stage, although orthotropic modeling significantly mitigates large overhanging areas, local violations—such as pore ceilings and tunnel intersections—may still persist. Therefore, we apply a continuous density-field optimization integrated with a differentiable layer-wise AM filter. This final step ensures self-supporting manufacturability while minimizing geometric deviation, resulting in a complex stochastic structure fully ready for direct fabrication.

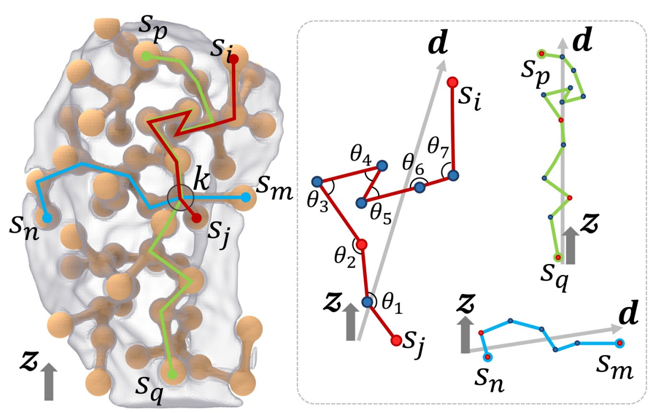


Figure 3: Graph abstraction and VP evaluation. The porous structure is represented as a connectivity graph where Gaussian kernels serve as nodes and tunnels as edges. For a given kernel k, VP is determined by analyzing the shortest topological paths connecting surface nodes (e.g., $s_i \to k \to s_j$) that pass through it. The final metric integrates path straightness (θ), total length, spatial depth, and the principal direction $\mathbf{d}$ (extracted via PCA).

3.2. Visual Permeability Metric

To parametrically quantify the concept of "visual permeability" lacking in prior structural optimization works, we abstract the pore system as a connectivity graph. As visually abstracted in Figure 3, each pore is represented as a node and each tunnel as an edge. Within this graph, a pathway $\mathcal{P}$ is defined as a sequence of connected pores forming a continuous hollow route through the structure, typically linking surface-adjacent pores. These pathways correspond to potential visual sightlines traversing the interior void space.

Under this abstraction, VP is governed by the structural properties of pathways rather than by individual pores or tunnels. Accordingly, we formulate a quantitative metric based on three key topological observations:

1) **Pathway Topology**: The internal void space can be viewed as a network of interlaced pathways, each traversing multiple pores and terminating near the surface.

2) **Straightness**: Straighter pathways exhibit higher VP, as they provide clearer and less occluded visual penetration.

3) **Depth Awareness**: Given comparable straightness, pathways with greater penetration depth contribute more significantly to perceived openness.

These observations provide a principled criterion for guiding connectivity optimization. We define the global VP of the structure, denoted as $\mathrm{VP}_{\mathrm{structure}}$, as the average permeability of its constituent kernels:

$$\mathrm{VP}_{\mathrm{structure}} = \frac{1}{N}\sum_{k=1}^{N} \mathrm{VP}_{\mathrm{kernel}}(k), \tag{1}$$

where N is the total number of Gaussian kernels. For an individual kernel k, its local permeability $\mathrm{VP}_{\mathrm{kernel}}(k)$ is determined by the optimal visual pathway traversing it:

$$\mathrm{VP}_{\mathrm{kernel}}(k) = \max_{s_i, s_j \in S_{\mathrm{surf}}} \left(\mathrm{VP}_{\mathrm{path}}(\mathcal{P}_{s_i \to k \to s_j})\right). \tag{2}$$

Let S be the set of all sampled Gaussian kernels, partitioned into a surface kernel set S_{surf} and an internal kernel set S_{int}. Here, $\mathcal{P}_{s_i \to k \to s_j}$ denotes the shortest topological path connecting two surface nodes $s_i, s_j \in S_{\mathrm{surf}}$ that passes through kernel k. Thus, for every kernel k, we evaluate all potential pathways passing through it, and the maximum score determines $\mathrm{VP}_{\mathrm{kernel}}(k)$.

The VP of a single path, $\mathrm{VP}_{\mathrm{path}}(\mathcal{P})$, is calculated by balancing four geometric components:

$$\begin{aligned}\mathrm{VP}_{\mathrm{path}}(\mathcal{P}) = w_{\mathrm{ang}} \cdot \mathrm{Ang}(\mathcal{P}) + w_{\mathrm{len}} \cdot \mathrm{Len}(\mathcal{P}) \\ + w_{\mathrm{loc}} \cdot \mathrm{Loc}(\mathcal{P}) + w_{\mathrm{dir}} \cdot \mathrm{Dir}(\mathcal{P}),\end{aligned} \tag{3}$$

where $w_{\mathrm{ang}}, w_{\mathrm{len}}, w_{\mathrm{loc}}, w_{\mathrm{dir}}$ are user-defined weights such that their sum is 1. The four components are defined as follows:

Angle. $\mathrm{Ang}(\mathcal{P})$ measures the straightness of the path. It is the product of the normalized angular deviations $\theta_i \in [0, 1]$ between consecutive segments, raised to the power of a sensitivity parameter α:

$$\mathrm{Ang}(\mathcal{P}) = \left(\prod_{i=1}^{|\mathcal{P}|-2} \theta_i\right)^{\alpha}, \tag{4}$$

where $|\mathcal{P}|$ denotes the number of nodes in the path.

Length. $\mathrm{Len}(\mathcal{P})$ rewards longer paths that penetrate deeper into the volume, following an exponential saturation curve characterized by a reference length L_0:

$$\mathrm{Len}(\mathcal{P}) = 1 - \exp\left(-\frac{|\mathcal{P}| - 1}{L_0}\right). \tag{5}$$

Location. $\mathrm{Loc}(\mathcal{P})$ encourages paths that traverse the internal volume rather than skimming the surface. It is modeled using a sigmoid function based on the proportion of internal nodes (r_{int}) versus surface nodes (r_{surf}):

$$\mathrm{Loc}(\mathcal{P}) = \frac{1}{1 + \exp\left(-\beta(r_{\mathrm{int}} - r_{\mathrm{surf}})\right)}. \tag{6}$$

Here, β controls the steepness of the transition; we set $\beta = 8.0$ in our implementation.

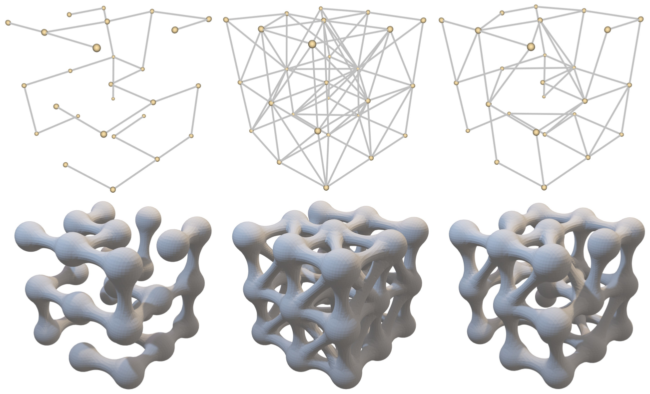
Direction. $\mathrm{Dir}(\mathcal{P})$ prioritizes horizontal VP, which is a preferred aesthetic in landscape stone design. To reliably capture the macroscopic orientation of a tortuous pathway, we apply principal component analysis (PCA) to the spatial coordinates of its constituent nodes. The principal eigenvector corresponding to the largest eigenvalue is extracted as the path's global trend direction $\mathbf{d}$. Let $\mathbf{z}$ be the vertical axis vector:

$$\mathrm{Dir}(\mathcal{P}) = 1 - |\mathbf{d} \cdot \mathbf{z}|. \tag{7}$$

With the metric defined, we can now evaluate and optimize the connectivity between pores.

3.3. Connectivity Optimization based on VP Metric

Given a fixed set of Gaussian kernels, our goal is to construct a pore connectivity graph that maximizes global VP while maintaining structural sparsity and manufacturability. Since indiscriminate edge addition leads to excessive fragmentation of the solid phase and poor printability, we formulate this task as a metric-driven, edge-budgeted connectivity optimization problem. As discussed in Section 2.5, existing routing algorithms are inadequate for aesthetic porous design: MST yields excessively sparse graphs that restrict visual pathways, and MSC generates dense, redundant connections that degrade manufacturability, as shown in Figure 4(a)(b). Therefore, we propose a customized **MBDST+** strategy.



$|E| = 26, VP = 0.639$ $|E| = 86, VP = 0.734$ $|E| = 50, VP = 0.721$

(a) MST (b) MSC (c) MBDST+

Figure 4: Visual and quantitative comparison of graph connectivity strategies. (a) The MST yields an excessively sparse graph with poor $VP = 0.639$. (b) The MSC achieves a marginally higher $VP = 0.734$, but at the cost of dense, redundant connections $|E| = 86$ that degrade manufacturability. (c) Our proposed **MBDST+** optimization achieves highly competitive $VP = 0.721$ while strictly bounding node degree and significantly reducing the total edge count $|E| = 50$ to preserve structural sparsity.

First, we construct an initial graph G_0 using a weighted MBDST [44, 13]. To prevent excessive local branching caused by dense tunnel intersections, which severely degrades manufacturability, a degree bound $\mathcal{D}_{max}$ is strictly imposed on each node. Furthermore, boundary-aware edge weights are incorporated to favor connections between internal and surface-adjacent pores. While this guarantees a sparse, manufacturable, and globally connected baseline, G_0 inherently provides limited VP.

To enhance perceptual openness, the connectivity is subsequently refined starting from G_0 through an iterative, kernel-centric optimization process. We define a candidate edge set E_{cand} consisting of spatially proximal pore pairs not directly connected in the initial tree. The optimization then proceeds iteratively in two stages:

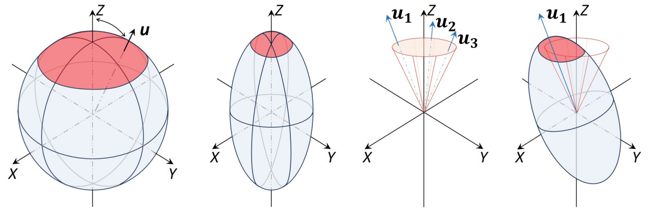
Edge addition. For each Gaussian kernel k, we evaluate all incident edges currently available in the candidate set E_{cand}. For each such edge $e = (k, j)$, we calculate the potential increment in global VP:

$$\Delta\text{VP}(e) = \text{VP}_{\text{structure}}(G \cup \{e\}) - \text{VP}_{\text{structure}}(G). \tag{8}$$

If there exists at least one candidate edge yielding $\Delta\text{VP}(e) > 0$, the edge providing the maximum gain is added to the graph, provided that the updated degree of kernel k does not exceed the prescribed bound $\mathcal{D}_{\max}$.

Edge replacement. If kernel k has already reached the maximum allowed degree $\mathcal{D}_{\max}$, direct edge addition is prohibited. In this scenario, we attempt to replace an existing incident edge of k with a candidate edge. For every potential replacement pair, we compute the resulting change in VP. A replacement is executed only if it simultaneously increases the local permeability contribution $\text{VP}_{\text{kernel}}(k)$ and yields a positive net gain in the global metric $\text{VP}_{\text{structure}}$. This ensures that local sightline improvements do not come at the expense of overall transparency, while strictly maintaining the node degree constraint.

The optimization sweeps over all kernels iteratively until no further local updates yield a positive global gain, or a predefined iteration limit is reached. As visually and quantitatively demonstrated in Figure 4(c), our optimized MBDST+ graph successfully achieves a highly competitive VP value while excellently controlling the number of edges. The final optimized graph serves as the topological skeleton for the subsequent implicit tunnel modeling.



(a) (b) (c) (d)

Figure 5: Gaussian kernel representations for pore generation. (a) Standard isotropic (spherical) kernels cause overhangs (red). (b) Orthotropic kernels elongated along the build axis ($\mathbf{z}$) reduce unsupported areas. (c) Orientation axes ($\mathbf{u}$) are sampled within a constrained cone to ensure printability. (d) Resulting geometry enhances natural irregularity while preventing increase support requirements.

3.4. Stochastic Porous Structure Modeling

Guided by the optimized MBDST+ skeleton from Section 3.3, this stage materializes the discrete graph into a continuous, support-free porous geometry. We build upon the Gaussian kernel-based implicit field [12], introducing specific fabrication-aware parameterizations to reduce overhangs.

3.4.1. Implicit Representation of Orthotropic Kernels

In the framework proposed by Tian et al. [12], each pore k is modeled by a Gaussian kernel:

$$G_k(\mathbf{p}) = \exp\left(-(\mathbf{p} - \mathbf{c}_k)^\intercal \Sigma_k^{-1} (\mathbf{p} - \mathbf{c}_k)\right), \tag{9}$$

where $\mathbf{c}_k$ denotes the center and Σ_k is the covariance matrix determining the kernel's spatial extent and orientation. While isotropic or arbitrarily anisotropic kernels facilitate the generation of organic morphologies, they inevitably introduce severe local overhangs during layer-by-layer fabrication, as highlighted by the red regions in Figure 5(a).

To improve printability, we restrict the kernels to an *orthotropic* configuration biased towards the build direction (the z-axis), as illustrated in Figure 5(b). In the local frame aligned with its principal orientation, the covariance is parameterized as:

$$\Sigma_{\text{local}} = \begin{bmatrix} \sigma_\perp^{-2} & 0 & 0 \\ 0 & \sigma_\perp^{-2} & 0 \\ 0 & 0 & \sigma_\parallel^{-2} \end{bmatrix}, \tag{10}$$

where $\sigma_\perp$ and $\sigma_\parallel$ control the transverse and axial extents, respectively. The world covariance is then obtained via $\Sigma = R\Sigma_{local}R^\intercal$, where R aligns the local z-axis to a sampled principal orientation vector $\mathbf{u}$.

3.4.2. Parameter Sampling and Overlap Handling

To reduce unsupported domes while maintaining organic randomness, the kernel parameters are carefully constrained:

Orthotropic scaling. The kernels are intentionally elongated along the build direction by sampling the transverse scale $\sigma_\perp \in [\sigma_{bound}, 2\sigma_{bound}]$ and defining the axial scale as $\sigma_\parallel = \rho\sigma_\perp$. The elongation ratio ρ is sampled within $(1, 1.5]$ to prevent generating unnaturally thin cavities.

Orientation restriction. Rather than uniform spherical sampling, the principal axis $\mathbf{u}$ is randomly sampled within a strict cone centered around the global build direction $\mathbf{z}$, as shown in

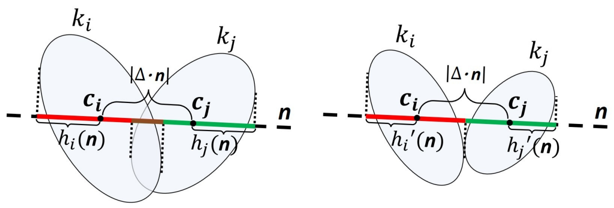


Figure 6: 2D illustration of the approximate Separating Axis Theorem (SAT). Overlapping kernels are efficiently detected by projecting their half-extents (h) onto a finite set of axes ($\mathbf{n}$) and are adaptively scaled down.

Figure 5(c)(d). The polar angle θ is constrained by a maximum deviation θ_{max} (set to 15°), balancing stochasticity with support-free necessity.

Overlap handling via SAT. Although kernel centers are initialized with a safety distance, orthotropic elongation may introduce overlaps at the target isosurface level τ. We efficiently detect overlaps using the separating axis theorem (SAT), as shown in Figure 6. For a pair of kernels i and j separated by $\Delta = \mathbf{c}_j - \mathbf{c}_i$, we evaluate a finite set of candidate axes $\mathcal{A}$ (comprising the center-to-center vector and their principal axes). The projected half-extent of kernel i along an axis $\mathbf{n} \in \mathcal{A}$ is:

$$h_i(\mathbf{n}) = \sqrt{-2\ln(\tau)\mathbf{n}^\intercal \Sigma_i \mathbf{n}}. \tag{11}$$

The two kernels are strictly separated if there exists any axis $\mathbf{n}$ satisfying:

$$|\Delta \cdot \mathbf{n}| > h_i(\mathbf{n}) + h_j(\mathbf{n}). \tag{12}$$

If overlapping, both kernels are uniformly scaled down based on the most deeply penetrating axis, avoiding invasive center relocation.

3.4.3. *Implicit Tunnel Modeling*

To materialize the connectivity, implicit tunnel fields are constructed precisely along the edges of the optimized MBDST+ graph. For an edge connecting kernels k_i and k_j, traditional cylindrical tubes would introduce significant lateral overhangs. Instead, the tunnel's cross-section is designed as an orthogonal ellipse derived from the endpoint metrics.

The implicit tunnel field is defined as:

$$T(\mathbf{p}) = \exp\left(-\frac{1}{2}(\mathbf{p}-\mathbf{s})_\perp^\intercal \Sigma_\perp^{-1} (\mathbf{p}-\mathbf{s})_\perp\right), \tag{13}$$

where $\mathbf{s}$ is the closest point projection of $\mathbf{p}$ onto the centerline $\mathbf{c}_i\mathbf{c}_j$, and $(\mathbf{p}-\mathbf{s})_\perp$ is the orthogonal distance vector. The 2D covariance matrix $\Sigma_\perp$ is obtained by projecting and averaging the 3D covariances (Σ_i and Σ_j) onto the transverse plane, naturally forming a support-friendly elliptical transition. Finally, the tunnel fields are additively blended with the pore fields to extract a unified, continuous porous isosurface, as shown in Figure 7.

3.5. *Optimization for Support-Free Structures*

Given the continuous stochastic porous geometry obtained from the previous stages, the final step aims to enforce strict support-free manufacturability while preserving the intended organic morphology. We formulate this task as a constrained density-field optimization problem on a voxelized representation. Rather than relying on heuristic geometric post-processing, additive manufacturing constraints are intrinsically embedded into the optimization process through a differentiable layer-wise operator.

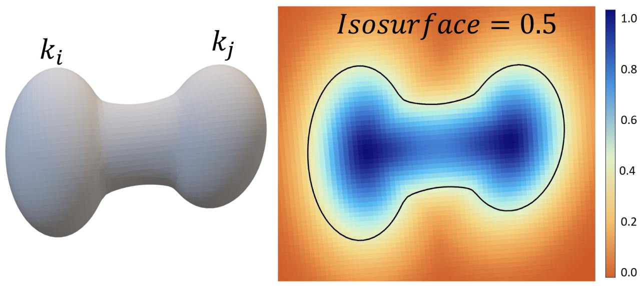


Figure 7: Implicit tunnel construction based on the graph topology. (Left) Two adjacent orthotropic kernels (k_i and k_j) connected along the optimized MBDST+ edge. (Right) The extracted isosurface exhibits a smooth, support-friendly transition tunnel with an elliptical cross-section seamlessly interpolating the endpoint metrics.

3.5.1. *Problem Formulation with Layer-Wise AM Filter*

Let $\mathbf{x} \in [0,1]^{|\Omega|}$ denote the independent design variables defined over the voxel domain Ω. After applying a standard Gaussian smoothing filter and a smooth Heaviside projection, we obtain the physical density field, denoted by $\bar{\rho}$.

To inherently model the layer-by-layer material deposition of 3D printing, we introduce a recursive layer-wise AM Filter that maps the physical density $\bar{\rho}$ to a printable density field ρ^P. Assuming the build direction is aligned with the z-axis, the printable density at layer k is recursively defined as:

$$\rho_k^P = S\left(\bar{\rho}_k, \mathcal{D}(\rho_{k-1}^P)\right), \quad k = 1, \ldots, N_z, \tag{14}$$

where $\mathcal{D}(\cdot)$ denotes a morphological dilation operator (implemented via 3×3 max-pooling) that models the geometric support domain provided by the underlying layer. The base layer ρ_0^P is initialized as fully supported by the build plate. To ensure the gradients can flow effectively during optimization, $S(a,b)$ is formulated as an analytical differentiable SoftMin operator:

$$S(a,b) = \frac{1}{2}\left(a + b - \sqrt{(a-b)^2 + \epsilon}\right). \tag{15}$$

This construction guarantees that a voxel can only be materialized if it is both structurally requested by the design field ($\bar{\rho}_k$) and sufficiently supported by the layer below ($\mathcal{D}(\rho_{k-1}^P)$).

The optimization objective is to minimize the geometric deviation between the printable structure ρ^P and the target stochastic field ρ^*, while rigorously restricting the total volume of unprintable (overhanging) material. The problem is formulated as:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathcal{J}(\rho^P, \rho^*) = \lambda_1 \mathcal{L}_{Dice}(\rho^P, \rho^*) + \lambda_2 \mathcal{L}_{MSE}(\rho^P, \rho^*), \\ \text{s.t.} \quad & G_{ov}(\mathbf{x}) = \|\bar{\rho} - \rho^P\|_1 \le \epsilon_{tol}, \\ & 0 \le \mathbf{x} \le 1, \end{aligned} \tag{16}$$

where $\mathcal{L}_{Dice}$ is a soft Dice loss that maximizes topological structure overlap, and $\mathcal{L}_{MSE}$ is a weighted mean squared error (MSE) penalizing local voxel deviations. The constraint G_{ov} precisely quantifies the total volume of unsupported material by calculating the L_1 norm of the difference between the intended physical density and the actual printable density.

3.5.2. *Sensitivity Analysis and Gradient Propagation*

The constrained optimization problem is solved using the method of moving asymptotes (MMA) [45]. This gradient-based solver requires the sensitivities of both the objective function and the overhang constraint with respect to the design variables $\mathbf{x}$.

For notational convenience, let $\mathcal{F} \in \{\mathcal{J}, G_{ov}\}$ represent a generic functional. The global sensitivity is derived via the chain rule, tracing back through the projection, filtering, and AM simulation stages:

$$\frac{\partial \mathcal{F}}{\partial x_i} = \sum_{j \in \Omega} \frac{\partial \mathcal{F}}{\partial \rho_j^P} \frac{\partial \rho_j^P}{\partial \bar{\rho}_j} \frac{\partial \bar{\rho}_j}{\partial \tilde{\rho}_j} \frac{\partial \tilde{\rho}_j}{\partial x_i}, \tag{17}$$

where $\tilde{\rho}$ denotes the intermediate smoothed density field.

The primary mathematical challenge lies in differentiating through the recursive AM filter. Due to its layer-wise dependency, the gradient at layer k aggregates not only the local sensitivity but also the support demand back-propagated from the layer above:

$$\frac{\partial \mathcal{F}}{\partial \rho_k^P} = \frac{\partial \mathcal{F}_{local}}{\partial \rho_k^P} + \underbrace{\frac{\partial \mathcal{F}}{\partial \rho_{k+1}^P} \frac{\partial S}{\partial b} \frac{\partial \mathcal{D}}{\partial \rho_k^P}}_{\text{Recursive Support Demand}} . \tag{18}$$

This recursive back-propagation explicitly encodes the physical mechanics of additive manufacturing: unsupported voxels in upper layers generate gradient signals that flow downward, compelling the optimizer to naturally grow the minimal necessary support structures from below. All operators involved in the AM filter are differentiable, and the resulting gradients are efficiently evaluated using automatic differentiation. The assembled sensitivities $\nabla \mathcal{J}(\mathbf{x})$ and $\nabla \mathcal{G}_{\text{ov}}(\mathbf{x})$ are then passed to the MMA solver to iteratively update the design variables until convergence.

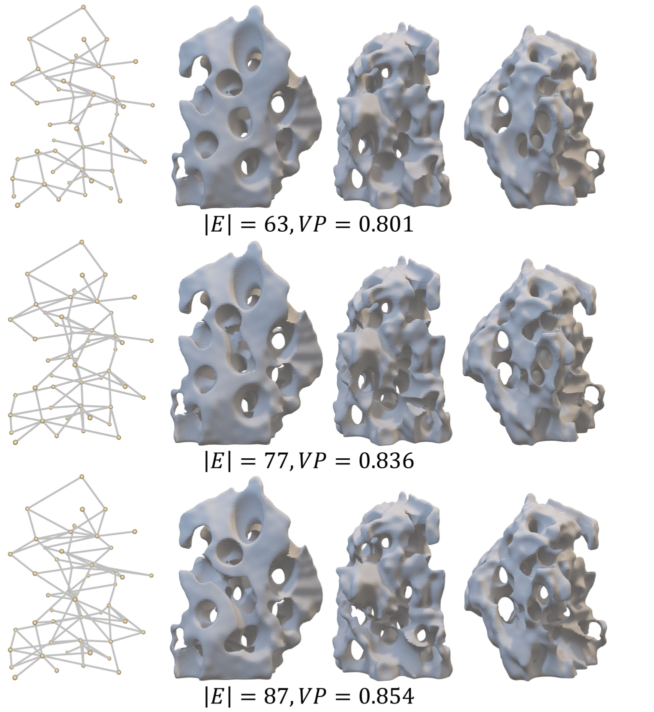


Figure 8: Visual validation of the proposed permeability metric. From top to bottom, the generated porous structures exhibit increasing VP scores (VP = 0.801, 0.836, 0.854) along with a corresponding increase in the number of edges ($|E| = 63, 77, 87$). As quantitatively reflected by the metric, higher VP scores strictly correspond to straighter, deeper, and more spatially coherent internal pathways, validating its alignment with human aesthetic perception.

4. Results and Discussion

In this section, we evaluate the proposed framework from three perspectives: the effectiveness of the visual permeability metric, the rigorous enforcement of manufacturability constraints, and the physical fabrication of our results using standard additive manufacturing processes.

4.1. Implementation Details

The proposed algorithms were implemented in C++ with CUDA acceleration for computationally intensive field evaluations. All experiments were conducted on a PC equipped with an Intel Core i9-9900K CPU @ 3.6 GHz and an NVIDIA GeForce RTX 2080 GPU. The default parameters in this paper were set as follows: the grid resolution for voxelization was set between 128^3 and 256^3 depending on the model complexity; the number of Gaussian kernels N ranged from 30 to 200; and the orthotropic elongation ratio ρ was sampled within (1.0, 1.5]. The weights for the visual permeability metric in Equation 3 were empirically set to $w_{ang} = 0.5$, $w_{len} = 0.1$, $w_{loc} = 0.1$, and $w_{dir} = 0.3$.

Regarding computational performance, the time complexity varies across the different stages of our decoupled pipeline. During the iterative graph connectivity optimization, the computational cost is independent of the voxel resolution and is instead bounded by $O(N^3)$, where N is the number of Gaussian kernels. In our experiments, a single connectivity iteration takes approximately 60 seconds for $N = 80$, and scales to about 900 seconds for $N = 200$.

Conversely, the implicit field generation and the layer-wise support-free optimization (SFOpt) scale linearly with the volumetric grid resolution, $O(|\Omega|)$. Generating the continuous SDF takes about 60 seconds at a 128^3 resolution and increases to over 500 seconds at 256^3. For the final density-field optimization, a single iteration takes merely 0.5 seconds at 128^3 and 4 seconds at 256^3, typically achieving full convergence within 80 to 200 iterations. As a representative benchmark, generating a complete porous structure configured with $N = 80$ kernels at a 128^3 resolution, comprising 3 iterations of connectivity optimization, continuous SDF generation, and 120 iterations of SFOpt, requires a total processing time of approximately 270 seconds.

4.2. Evaluation of Visual Permeability

A core contribution of our work is the quantitative enhancement of visual permeability. To validate the effectiveness of our VP metric and the graph-based optimization strategy, we conducted a comparative study on connectivity generation methods.

4.2.1. Metric Validation

We first analyze the correlation between the computed VP scores and the perceptual quality of the structures. As illustrated in Figure 8, we generated a sequence of porous structures with increasing VP scores. It can be observed that structures with lower scores exhibit cluttered internal layouts with short, tortuous tunnels that obstruct visual penetration. In contrast, structures with higher scores feature straighter, deeper, and more spatially coherent pathways, which aligns with the aesthetic preference in landscape porous artifacts.

4.2.2. Efficacy of Connectivity Optimization

We compared our MBDST+ connectivity strategy against two established baseline methods: (1) MST: A Minimum Spanning Tree based solely on Euclidean distance [42], representing the

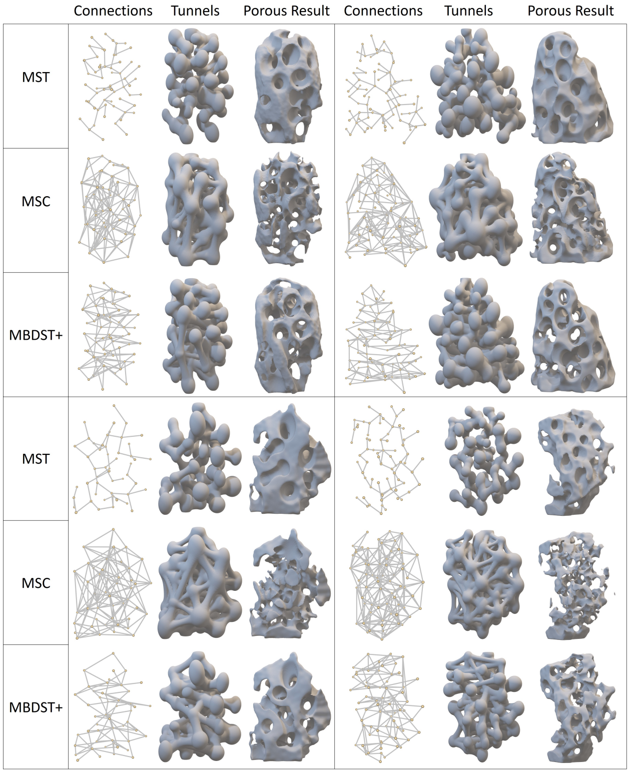
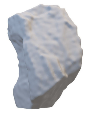


Figure 9: Visual comparison of porous structures driven by different connectivity strategies. MST results in overly sparse skeletons that occlude visual pathways. MSC generates dense, redundant topologies that exacerbate manufacturability issues. Our MBDST+ achieves an optimal aesthetic balance, providing highly transparent pathways with controlled structural complexity.

Table 1: Quantitative comparison of connectivity strategies (MST, MSC, MBDST+) and the impact of the final SFOpt across varying kernel counts N. Our proposed MBDST+ strategy consistently yields the highest VP while maintaining a bounded edge degree. Notably, the SFOpt stage minimizes unsupported volume to negligible levels without compromising the optimized VP or porosity.

Model	Type	Edges	Porosity	VP	Unsup Voxels
 $N = 60$	MST	59	40.68%	0.6979	1102
	MSC	180	66.75%	0.7491	619
	MBDST+	112	58.24%	0.8521	1008
	SFOpt	112	51.78%	0.8521	6
 $N = 80$	MST	79	40.69%	0.6735	749
	MSC	200	63.59%	0.7345	1518
	MBDST+	150	56.67%	0.8724	1732
	SFOpt	150	50.85%	0.8724	7
 $N = 50$	MST	49	41.04%	0.7177	558
	MSC	162	68.71%	0.7691	1467
	MBDST+	89	54.15%	0.8534	1110
	SFOpt	89	48.47%	0.8534	5
 $N = 70$	MST	69	40.36%	0.6702	486
	MSC	260	70.89%	0.7839	1060
	MBDST+	140	61.48%	0.8572	929
	SFOpt	140	53.74%	0.8572	12

lower bound of connectivity. (2) MSC: A topological skeleton extraction method commonly used in implicit porous modeling [12], representing a density-driven baseline.

As detailed in Table 1 and visually compared in Figure 9, the different strategies yield starkly contrasting outcomes. The MST approach ensures structural sparsity but completely fails to establish meaningful, straight visual pathways, resulting in the lowest VP scores across all configurations. Conversely, the MSC method aggressively extracts all topological ridges; while it achieves a marginally higher VP, it comes at the severe cost of geometric redundancy. This massive edge count clutters the interior space and severely degrades initial manufacturability.

Our proposed MBDST+ strategy strikes an effective balance between connectivity and complexity. By optimizing for pathway straightness and depth while strictly constraining the local node degree, it consistently achieves the highest VP scores. For instance, the model in the second row ($N = 80$) yields a VP = 0.8724, outperforming MST by over 29% and MSC by 18%, while maintaining a concise topology with only 150 edges. This confirms that our optimization strategy successfully embeds aesthetic intent into the topological construction phase without sacrificing sparsity.

Furthermore, Table 1 highlights the indispensable role of our SFOpt. Across all kernel configurations, after the layer-wise AM filter is applied, the unsupported voxel count is drastically reduced to near zero. Crucially, this rigorous manufacturability enforcement preserves the exact VP scores and edge counts achieved by the MBDST+ graph, proving that our framework successfully eliminates internal overhangs with minimal geometric distortion to the aesthetic design.

4.3. Geometric Quality and Manufacturability

To demonstrate the robustness and generalization of our generative framework, we applied our pipeline to a diverse set of input geometric domains with varying kernel counts N. As shown in Figure 10, our method consistently produces highly organic, biomimetic porous morphologies that adapt naturally to the global shape of the input bounds.

Figure 10 visually validates the efficacy of our continuous density-field optimization. Before the SFOpt, while the orthotropic kernels mitigate massive overhangs, local unprintable violations, such as pore domes and horizontal tunnel ceilings, inevitably persist. After applying our differentiable layer-wise AM filter, the optimizer intelligently eliminates these internal overhangs by carving out problematic regions or by seamlessly growing upward compensatory supports, without requiring invasive global deformation. The resulting structures achieve nearly 100% support-free capability while faithfully preserving the stochastic aesthetics and visual permeability dictated by the initial MBDST+ graph.

4.4. Physical Fabrication

To demonstrate the practical applicability and true manufacturability of our framework, we physically fabricated several generated samples using fused deposition modeling (FDM) on a Bambu Lab X1C 3D printer with PLA filament.

As showcased in Figure 11, all models were printed exactly as-is, directly from the generated geometry without the addition of any automatically generated internal supports from the slicing software. The successful prints confirm that our layer-wise optimization rigorously satisfies the maximum overhang angle constraints. Furthermore, the back-lit photographs clearly highlight that the internal channels are entirely hollow, clean, and spatially continuous. The light effortlessly penetrates the complex

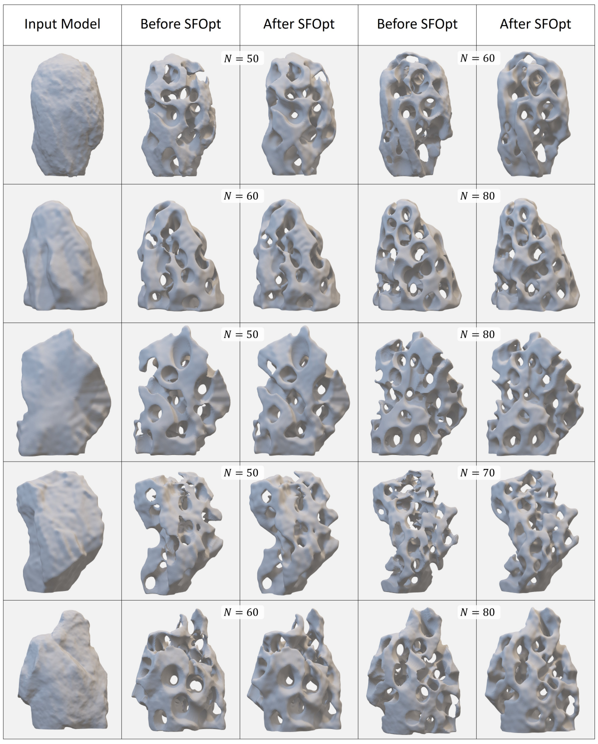


Figure 10: Generalization of our generative framework across diverse input models and varying kernel counts N. For each configuration, the structure before and after the density-field SFOpt is shown. The optimization successfully eliminates local internal overhangs (ensuring printability) while faithfully preserving the original stochastic morphology and visual permeability.

stochastic voids, validating both the robust support-free capability and the high-fidelity aesthetic visual permeability achieved by our computational designs.

5. Conclusion

In this paper, we presented a comprehensive, fabrication-oriented generative framework for designing support-free stochastic porous structures. Recognizing the critical gap between procedural aesthetic modeling and the physical constraints of additive manufacturing, we introduced a unified pipeline driven by both perceptual aesthetics and strict printability. First, we formulated a quantitative visual permeability metric that explicitly parameterizes the traditional aesthetic concept of transparency, utilizing it to drive a customized MBDST+ graph optimization. Next, we implicitly modeled the continuous porous geometry using orthotropic Gaussian kernels, which intrinsically mitigates initial overhangs without sacrificing organic smoothness. Finally, we enforced rigorous global printability through a continuous density-field optimization governed by a differentiable layer-wise AM filter. Extensive experiments and physical fabrications confirm that our method successfully resolves the fundamental trilemma of maintaining organic stochasticity, enabling explicit aesthetic control, and guaranteeing support-free manufacturability.

Limitations. While our framework effectively generates fabricable porous structures, it has several limitations. Because the connectivity optimization relies on a fixed set of randomly initialized kernels, the maximum achievable visual permeability is inherently bounded by this initial sampling. Additionally, our current VP metric prioritizes topological sightlines but does not explicitly account for the volumetric cross-sectional area of the pores. To guarantee printability, the orthotropic kernels are strictly elongated along the global z-axis; this hard constraint inherently biases the organic aesthetic and limits morphological diversity. From a fabrication standpoint, minor filament drooping can still occur in FDM-printed samples, because our AM filter ensures geometric overhang compliance but ignores the complex thermomechanical behaviors of molten polymers. Finally, our dense voxel optimization scales cubically $O(|\Omega|)$, which currently poses a memory bottleneck for scaling up to architectural-level designs.

Future Work. These limitations open exciting avenues for future research. To transcend the current aesthetic bounds, future work could formulate a joint optimization that simultaneously updates both topological connectivity and kernel spatial coordinates, alongside incorporating localized cross-sectional sizing into the VP metric. Furthermore, the morphological richness could be significantly expanded by decoupling the visual pathways from the manufacturing build direction, potentially leveraging multi-axis additive manufacturing.

References

[1] S. J. Hollister, Porous scaffold design for tissue engineering, Nature materials 4 (7) (2005) 518–524. `doi:10.1038/nmat1421`.

[2] P. X. Ma, Biomimetic materials for tissue engineering, Advanced drug delivery reviews 60 (2) (2008) 184–198. `doi:10.1016/j.addr.2007.08.041`.

[3] C. M. Parlett, K. Wilson, A. F. Lee, Hierarchical porous materials: catalytic applications, Chemical Society Reviews 42 (9) (2013) 3876–3893. `doi:10.1039/c2cs35378d`.

[4] W. Zhang, H. Pan, L. Lu, X. Duan, X. Yan, R. Wang, Q. Du, Dualms: Implicit dual-channel minimal surface optimization for heat exchanger design, in: Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers, 2025, pp. 1–10.

[5] M. Bernhard, M. Bolhassani, M. Akbarzadeh, Performative porosity–adaptive infills for architectural elements, in: Proceedings of IASS Annual Symposia, Vol. 2020, International Association for Shell and Spatial Structures (IASS), 2020, pp. 1–13.

[6] Z. Feng, P. Gu, M. Zheng, X. Yan, D. Bao, Environmental data-driven performance-based topological optimisation for morphology evolution of artificial taihu stone, in: The international conference on computational design and robotic fabrication, Springer, 2021, pp. 117–128.

[7] Q. Deng, X. Li, Y. Liu, K. Hu, Exploration of three-dimensional spatial learning approach based on machine learning–taking taihu stone as an example, Architectural Intelligence 2 (1) (2023) 5.

[8] M. P. Bendsøe, N. Kikuchi, Generating optimal topologies in structural design using a homogenization method, Computer methods in applied mechanics and engineering 71 (2) (1988) 197–224.

[9] A. T. Gaynor, J. K. Guest, Topology optimization considering overhang constraints: Eliminating sacrificial support material in additive manufacturing through design, Structural and Multidisciplinary Optimization 54 (5) (2016) 1157–1172.

[10] M. Langelaar, An additive manufacturing filter for topology optimization of print-ready designs, Structural and multidisciplinary optimization 55 (3) (2017) 871–883.

[11] X. Yan, C. Rao, L. Lu, A. Sharf, H. Zhao, B. Chen, Strong 3d printing by tpms injection, IEEE Transactions on Visualization and Computer Graphics 26 (10) (2019) 3037–3050.

[12] L. Tian, L. Lu, W. Chen, Y. Xia, C. C. Wang, W. Wang, Organic open-cell porous structure modeling, in: Proceedings of the 5th Annual ACM Symposium on Computational Fabrication, 2020, pp. 1–12.

[13] L. Tian, Z. Wang, X. Liu, K. Tian, A. Sharf, L. Lu, Fabricable stochastic periodic porous microstructures: A wang cube and gaussian kernel approach, Additive Manufacturing 103 (2025) 104739.

[14] W. Wang, Y.-J. Liu, J. Wu, S. Tian, C. C. Wang, L. Liu, X. Liu, Support-free hollowing, IEEE transactions on visualization and computer graphics 24 (10) (2017) 2787–2798.

[15] M. P. Bendsøe, Optimal shape design as a material distribution problem, Structural optimization 1 (4) (1989) 193–202.

Figure 11: Physical fabrication results. FDM printed samples (PLA) printed without internal supports, demonstrating support-free geometry. Back-lit photography highlighting the visual permeability and continuous internal voids characteristic of these porous structures.

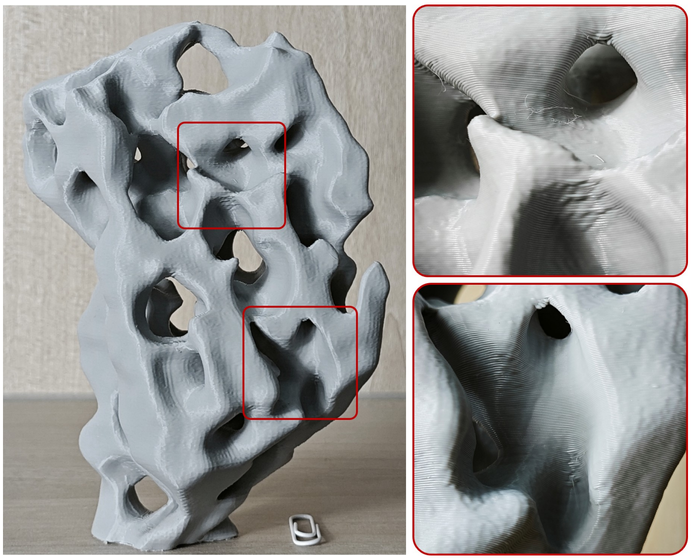


Figure 12: Physical fabrication results and artifact analysis. (Left) The FDM printed sample demonstrates overall successful support-free geometry. (Right) In the red boxes, minor physical artifacts, such as slight filament drooping and separation, are observable in localized horizontal bridging areas. These imperfections highlight the gap between theoretical geometric overhang constraints and the complex thermomechanical realities of FDM extrusion.

[16] X. Qian, Undercut and overhang angle control in topology optimization: a density gradient based integral approach, International Journal for Numerical Methods in Engineering 111 (3) (2017) 247–272.

[17] A. Garaigordobil, R. Ansola, J. Santamaría, I. Fernández de Bustos, A new overhang constraint for topology optimization of self-supporting structures in additive manufacturing, Structural and Multidisciplinary Optimization 58 (5) (2018) 2003–2017.

[18] E. van de Ven, R. Maas, C. Ayas, M. Langelaar, F. van Keulen, Continuous front propagation-based overhang control for topology optimization with additive manufacturing, Structural and Multidisciplinary Optimization 57 (5) (2018) 2075–2091.

[19] E. van de Ven, R. Maas, C. Ayas, M. Langelaar, F. van Keulen, Overhang control in topology optimization: a comparison of continuous front propagation-based and discrete layer-by-layer overhang control, Structural and Multidisciplinary Optimization 64 (2) (2021) 761–778.

[20] Q. Cui, T. Langlois, P. Sen, T. Kim, Fast and robust stochastic structural optimization, in: Computer Graphics Forum, Vol. 39, Wiley Online Library, 2020, pp. 385–397.

[21] O. Al-Ketan, D.-W. Lee, R. K. A. Al-Rub, Mechanical properties of additively-manufactured sheet-based gyroidal stochastic cellular materials, Additive Manufacturing 48 (2021) 102418.

[22] L. Tian, B. Sun, X. Yan, A. Sharf, C. Tu, L. Lu, Continuous transitions of triply periodic minimal surfaces, Additive Manufacturing 84 (2024) 104105.

[23] X. Liu, C. Li, L. Lu, O. Deussen, C. Tu, Fabricable multi-scale Wang tiles, Computer Graphics Forum 41 (5) (2022) 149–159. `doi:10.1111/cgf.14610`.

[24] Y. Xu, H. Pan, R. Wang, Q. Du, L. Lu, New families of triply periodic minimal surface-like shell lattices, Additive Manufacturing 77 (2023) 103779.

[25] T. Xue, L. Liu, L. Lu, P. Henderson, P. Tang, H. Li, J. Liu, H. Zhao, H. Peng, B. Bickel, Mind: Microstructure inverse design with generative hybrid neural representation, in: Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Conference Papers, 2025, pp. 1–12.

[26] X. Kou, S. Tan, A simple and effective geometric representation for irregular porous structure modeling, Computer-Aided Design 42 (10) (2010) 930–941.

[27] L. Lu, A. Sharf, H. Zhao, Y. Wei, Q. Fan, X. Chen, Y. Savoye, C. Tu, D. Cohen-Or, B. Chen, Build-to-last: Strength to weight 3d printed objects, ACM Transactions on Graphics (ToG) 33 (4) (2014) 1–10.

[28] J. Martínez, J. Dumas, S. Lefebvre, Procedural voronoi foams for additive manufacturing, ACM Transactions on Graphics (TOG) 35 (4) (2016) 1–12.

[29] A. Roberts, E. J. Garboczi, Elastic properties of model random three-dimensional open-cell solids, Journal of the Mechanics and Physics of Solids 50 (1) (2002) 33–55.

[30] J.-H. Groth, C. Anderson, M. Magnini, C. Tuck, A. Clare, Five simple tools for stochastic lattice creation, Additive Manufacturing 49 (2022) 102488.

[31] A. S. Ullah, D. M. D'Addona, Y. Seto, S. Yonehara, A. Kubo, Utilizing fractals for modeling and 3d printing of porous structures, Fractal and Fractional 5 (2) (2021) 40.

[32] A. Sharif Ullah, Design for additive manufacturing of porous structures using stochastic point-cloud: a pragmatic approach, Computer-Aided Design and Applications 15 (1) (2018) 138–146.

[33] A. S. Ullah, H. Kiuno, A. Kubo, D. M. D'Addona, A system for designing and 3d printing of porous structures, CIRP Annals 69 (1) (2020) 113–116.

[34] X. Shi, D. Li, J. Chen, Y. Chen, 3d-porous-gan: a high-performance 3d gan for digital core reconstruction from a single 3d image, Journal of Petroleum Exploration and Production Technology 13 (12) (2023) 2329–2345.

[35] K. Naghavi Khanghah, Z. Wang, H. Xu, Reconstruction and generation of porous metamaterial units via variational graph autoencoder and large language model, Journal of Computing and Information Science in Engineering 25 (2) (2025) 021003.

[36] J. Martínez, S. Hornus, H. Song, S. Lefebvre, Polyhedral voronoi diagrams

for additive manufacturing, ACM Transactions on Graphics (TOG) 37 (4) (2018) 1–15.

[37] B. Ezair, F. Massarwi, G. Elber, Orientation analysis of 3d objects toward minimal support volume in 3d-printing, Computers & Graphics 51 (2015) 117–124.

[38] N. Umetani, R. M. Schmidt, Cross-sectional structural analysis for 3d printing optimization., in: SIGGRAPH Asia technical briefs, 2013, pp. 5–1.

[39] R. Prévost, E. Whiting, S. Lefebvre, O. Sorkine-Hornung, Make it stand: balancing shapes for 3d fabrication, ACM Transactions on Graphics (ToG) 32 (4) (2013) 1–10.

[40] J. Etienne, N. Ray, D. Panozzo, S. Hornus, C. C. Wang, J. Martínez, S. McMains, M. Alexa, B. Wyvill, S. Lefebvre, Curvislicer: Slightly curved slicing for 3-axis printers, ACM Transactions on Graphics (TOG) 38 (4) (2019) 1–11.

[41] C. Dai, C. C. Wang, C. Wu, S. Lefebvre, G. Fang, Y.-J. Liu, Support-free volume printing by multi-axis motion, ACM Transactions on Graphics (TOG) 37 (4) (2018) 1–14.

[42] S. Pettie, V. Ramachandran, An optimal minimum spanning tree algorithm, Journal of the ACM (JACM) 49 (1) (2002) 16–34.

[43] H. Edelsbrunner, J. Harer, V. Natarajan, V. Pascucci, Morse-smale complexes for piecewise linear 3-manifolds, in: Proceedings of the nineteenth annual symposium on Computational geometry, 2003, pp. 361–370.

[44] M. Singh, L. C. Lau, Approximating minimum bounded degree spanning trees to within one of optimal, Journal of the ACM (JACM) 62 (1) (2015) 1–19. doi:10.1145/2629366.

[45] K. Svanberg, The method of moving asymptotes—a new method for structural optimization, International journal for numerical methods in engineering 24 (2) (1987) 359–373.